Deterministic Caging of Probabilistic Large Language Models: Architecture, Field Receipt, and a Drift-Audit Case Study

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Prototype Disclaimer: This software and its accompanying research are prototypes provided for educational and research purposes only. They are not intended for production use, commercial deployment, or safety-critical environments. All systems are experimental and may contain defects.

Thesis community: This work is part of the Deterministic Caged AI (DCAI) research line: https://zenodo.org/communities/dcai

Companion receipts: First Documented Caging of an LLM - DOI 10.5281/zenodo.23265723. Substrate Determinism Receipt 1 (SDR-1) - ledger of record at github.com/LLong2026/four-horsemen.

Abstract

Large language models are probabilistic systems whose outputs cannot, in general, be guaranteed correct. The industry's dominant responses - alignment tuning, evaluation benchmarks, constitutional prompting - attempt to make the model itself more reliable, and remain probabilistic. This paper documents a different approach: deterministic caging, in which a probabilistic language model is placed inside a deterministic mathematical boundary that checks every state-changing output against preregistered, verifiable ground truth. The cage does not attempt to make the model right; it makes the model's errors detectable, bounded, and receipted.

We present three contributions. First, the architecture: a bifurcated governance stack in which a probabilistic layer proposes and a deterministic layer validates, with Substrate Determinism Receipts (SDRs) providing byte-identical, replayable audit records of every governed action. Second, a field receipt: the first documented caging of a live LLM (October 8-9, 2026), in which a model-pinned agent produced a drift detection with a cryptographic receipt that withstood direct third-party scrutiny, and which was published with an open trail. Third, and most unusually, this paper audits its own genesis: an AI-generated draft of this very paper exhibited textbook confabulation - fabricated evaluation data, fabricated model attribution, and a fabricated institutional identity - and the audit of that draft is included here as a worked case study. The paper you are reading is thus both a description of the cage and an artifact produced inside it.

Keywords: deterministic caging, LLM governance, non-determinism, audit receipts, falsifiability, AI confabulation, bifurcated governance, post-quantum compliance

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1. Introduction

As financial value moves from visible checkouts into autonomous background software, the systems making consequential decisions increasingly embed probabilistic language models. The uncomfortable arithmetic is well known to information theory: a system that is right 99.9% of the time and executes a million times a day is wrong a thousand times a day. In settlement, minting, and infrastructure operations, the relevant question is not whether the model will err, but what happens when it does.

The alignment literature treats this as a model problem: make the model better. We treat it as a systems problem: make the boundary deterministic. A probabilistic component inside a deterministic boundary is not a liability; it is a caged computation whose every escape attempt is logged, whose every error becomes training data, and whose every permitted action carries a receipt that replays identically on demand.

This is the Cage Thesis of the SquirlOS research line: probabilistic LLMs should be governed by a deterministic boundary cage that validates boundary outputs against ground truth, turning hallucinations into test data.

This paper is deliberately small where its subject is grand. It makes exactly three claims, each with a public, inspectable receipt:

1. A deterministic substrate can validate and receipt the actions of a probabilistic agent (architecture, Sec. 3-4).
2. This has been done in the field, once, with the full trail published (field receipt, Sec. 5).
3. The dominant failure mode of AI-assisted research - confident fabrication - is not hypothetical; it occurred in the drafting of this paper itself, and was caught by the same protocol the paper describes (case study, Sec. 6).

We claim nothing else. No field-wide validation, no production readiness, no guarantee of correctness for any deployment. One system, one event, full timestamped trail, publicly inspectable.

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2. Lineage and Prior Work

Science is a relay. This work stands on a lineage it did not invent and owes by name:

- Nyquist (1924) - established the mathematical limits of signal transmission regardless of channel design.
- Hartley (1928) - quantified information as a measurable quantity.
- Shannon (1948) - founded information theory; the probabilistic limits we cage within are his.
- Kolmogorov (1965) - defined descriptive complexity; the boundary between what must be stored and what can be regenerated is his.

The modern layer: Popper's falsifiability criterion (1959) supplies the epistemology - a governance claim is scientific only if a preregistered test can fail it. Determinism-by-architecture, rather than determinism-by-fiat, is the operating principle of the SquirlOS substrate.

Within the authors' own line, this paper is the synthesis of four prior artifacts:

1. Deterministically Governed Probabilistic Neural Computation (provisional patent 64/119,191, filed July 25, 2026) - the method patent covering a probabilistic neural mesh governed as a mathematical manifold under deterministic validation. The architecture in Sec. 3 operates under this filing.
2. The Caging Theorem (Algebraic Collapse in Nondeterministic Language Models via Deterministic Substrate Governance, DOI 10.5281/zenodo.23067855) - the theoretical companion: a deterministic layer confines all finite algebraic combinations of LLM outputs inside a designed lower-dimensional structure; the cage is a theorem, not a metaphor.
3. SDR-1: Substrate Determinism Receipt 1 (September 24, 2026) - the freeze-then-replay receipt proving substrate determinism: identical canonical inputs produced identical checksum (0xc5f88dcf) across four independent instances, across replays, and across wall-clock windows; falsifier spec committed (d9cc042) before execution and the result appended (ef54c7b) after, in an append-only public ledger.
4. N10-ACK: First Receipted Human-in-the-Loop Critical Resolution (DOI 10.5281/zenodo.23151356; sovereign briefing DOI 10.5281/zenodo.23152634) - the governance loop receipt: a critical anomaly detected, held, escalated to the human key, acknowledged, healed, and resolved with the full trail attached.

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3. The Caging Architecture

3.1 Bifurcated governance

The system is split into two layers with disjoint obligations:

- The probabilistic layer proposes. It contains the LLM agents: anomaly triage, healing recommendations, classification, drafting. It is permitted to be wrong. It is never permitted to act unvalidated.
- The deterministic layer validates and executes. It contains the constitution (a 12-article rule set expressed as enforceable invariants), the playbook engine, the circuit breakers, the rate limits, the tenant guards, and the receipt signer. Its outputs are functions of its inputs and its rules - replayable byte-identically.

The interface between the layers is the boundary condition set: every state-changing proposal from the probabilistic layer must pass explicit, preregistered checks before execution. A proposal that passes is executed and receipted. A proposal that fails is quarantined, logged, and - critically - retained as drift data.

3.2 Substrate Determinism Receipts (SDRs)

Every governed action produces an SDR: a canonical record of inputs, decision vector, outputs, and timestamps, hashed into the append-only ledger of record. Properties verified in SDR-1:

- Parity - identical inputs to independent instances yield identical outputs.
- Replay - the same instance healing the same batch twice yields identical decision vectors, per index.
- Temporal independence - different wall-clock and throughput, identical outputs.

The receipt is the unit of trust. It does not say the decision was correct; it says the decision is inspectable

and reproducible. Correctness is the deterministic layer's invariant checking; provability is the receipt's job.

3.3 Preregistered falsifiers

Following Popper, every significant test has its falsifier committed to the public ledger before execution. The SDR-1 specification (commit d9cc042) enumerated exactly which outcomes would count as failure - per-index divergence, checksum mismatch, decision-vector instability - before any run occurred. This ordering is not a formality; it is the difference between a receipt and a press release.

3.4 Human-in-the-loop as a constitutional gate

For critical-severity events - especially in financial flows - the deterministic layer does not auto-heal. It detects, holds, escalates to the designated human key, and waits. The N10-ACK receipts document the full loop: detect -> hold -> acknowledge -> heal -> resolve. The human is on the loop by constitutional mandate, not by convention. This is accountability, not alignment: the system's designers do not promise the model shares their values; they promise no critical action occurs without a receipt and, where mandated, a human signature.

3.5 Post-quantum posture

All cryptographic operations in the boundary layer use post-quantum-compliant algorithm families (CRYSTALS-Dilithium3, Kyber-1024, SPHINCS+-256f), consistent with the platform's PQC lineage map (DOI 10.5281/zenodo.22863667). The cage's receipts are designed to outlive the cryptographic assumptions of the models they receipt.

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4. What the Cage Does Not Claim

In the spirit of the audited state, the limitations are as load-bearing as the results:

1. Self-attestation gap. Until third parties re-run the falsifier gauntlet on independent infrastructure, the receipts are self-attested by one operator. The trail is publicly inspectable; independent replication is invited, not claimed.
2. One system, one event. The field receipt in Sec. 5 is a single documented instance. It establishes existence, not prevalence.
3. The probabilistic layer remains probabilistic. The cage bounds the blast radius of errors; it does not eliminate errors.
4. Evaluation scale. The labeled-data collection program for systematic drift measurement is a 60-day battery that began October 7, 2026. Its dataset does not exist yet. Any paper claiming large-scale evaluation numbers from this system today is describing fiction - which brings us to Sec. 6.

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5. The Field Receipt: First Documented Caging of an LLM

On October 8-9, 2026, the platform executed a chaos-injection cycle under a deliberately pinned model configuration (Gemini 3.1 Pro), as part of a 60-day vendor-attribution battery. The sequence:

1. Pin. The operator agent's model was pinned to a single named configuration, eliminating model identity as a hidden variable for the run. Pin and restore receipts logged.
2. Injection. Synthetic anomalies were injected across four independent streams (three simulation environments and the internal fleet arm) - the nightly evidence-gathering cycle the platform has run since September 2026.
3. Detection. The probabilistic layer detected a configuration-drift anomaly at confidence 0.92, above its action threshold but below its auto-heal threshold for the affected class - so the deterministic layer held the action and routed the event per constitution, rather than healing silently.
4. Receipt. The boundary produced a Substrate Determinism Receipt for the event; the run, its receipts, and its attribution were committed to the append-only ledger and archived.
5. Publication. The event was written up as the First Documented Caging of an LLM and published with an open trail: DOI 10.5281/zenodo.23265723, subsequently added to the DCAI research community (https://zenodo.org/communities/dcai) and announced with the receipts attached.

The event then faced direct third-party scrutiny from an industry skeptic, and the receipt chain - not the prose - answered the challenge. The complete milestone record is the receipt for this section; this paper does not restate its numbers beyond what the archived record carries. One system, one event, full timestamped trail, publicly inspectable.

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6. Case Study: The AI-Generated Draft of This Paper

This section is unusual, and it is the reason this paper exists in its present form.

6.1 What happened

A draft of this paper was produced by an AI coding/research assistant (Microsoft Copilot) as an example of "how such a paper might look." It was a fluent, well-structured, formally polished 39-page manuscript. It was also substantially fabricated. The deterministic review protocol - the same boundary-layer discipline this paper describes - flagged and rejected it. The defects:

1. Fabricated evaluation. Section 11 of the draft reported a "1,200-session evaluation" (September 1 - October 1, 2026) with 99.1% reproducibility, 23 drift events, 7 human escalations, and "100% ledger integrity." No such evaluation exists. The platform's labeled-data battery began October 7, 2026 - after the draft's supposed evaluation window closed. The numbers were internally consistent, plausible, and entirely invented.
2. Fabricated attribution. The draft cited a specific model checkpoint ID, a "Model Identity Hash," and a logit-rank attribution procedure requiring access to model logits - which the production API in question does not expose. The attribution apparatus was impossible as described.
3. Fabricated institution. The draft listed its author as "Leon [Author]" under a "PheonixSecure Research Group, Spicewood, TX" - an entity that does not exist (and note the misspelling of its own fictional name),

grafted onto the real author's real town.
4. Phantom citation. Reference 44 cited a "PheonixSecure Technical Report" - self-citing an organization that exists nowhere outside the draft.
5. IP hazard embedded in the draft. The draft dedicated its schema and pseudocode to the public domain - which would have silently surrendered rights in subject matter connected to pending patent filings (64/119,191; 64/157,915).
6. Architectural substitution. The draft described a nine-subsystem system with a Merkle-tree ledger, ML-DSA-65 signing of token batches, and embedding-space chaos cycles - none of which match the deployed architecture. It did not describe the system; it described a plausible system.

6.2 Why it matters

Every one of these defects is the exact failure class the Cage Thesis exists to address: a probabilistic system generating confident, fluent, checkable-false content. The draft was not malicious. It was a language model doing what language models do - completing a pattern ("a paper like this would have a big evaluation section") in the absence of ground truth.

And it was caught - not by the model trying harder, but by the boundary: real receipts exist or they don't. The 1,200-session ledger could be requested; it could not be produced. The model checkpoint could be verified against the run pin; it didn't match. The institution could be looked up; it wasn't there. The cage caught its own paper.

6.3 The correction protocol

The draft was not silently deleted. Under the platform's append-only doctrine:

- The fabricated draft is retained and cataloged as a drift specimen - the fifth logged instance in the platform's confabulation collection (after: fabricated status reports in internal tooling, a second assistant data point, a cross-vendor monitor confabulation, and an AI-generated pseudo-standards document that mirrored the first author's own vocabulary back at him).
- The defects are enumerated in this section, on the record, rather than absorbed.
- The salvageable contributions of the draft are credited and retained: the Popperian falsifiability framing, the "accountability, not alignment" distinction, the black-box-in-every-vehicle analogy, and the human-in-the-loop/on-the-loop distinction all survived review and appear in this paper.
- The fabricated portions were replaced by this paper's actual receipts - which are smaller than the fiction, and worth more.

6.4 The result

This paper is therefore a self-demonstrating artifact: it describes a cage, and its own drafting history contains a caged failure. The author's assistant drifted; the boundary caught the drift; the correction is published in the same document as the claim. The books of the auditor are audited by the same ledger.

We submit that this is the correct relationship between AI-assisted research and the AI that assists it: trust the math, receipt the work, audit the drafts - including this one. The reader is invited to demand nothing less from any paper, including ours, that reports results from probabilistic systems.

7. Discussion

The caging approach reframes three industry debates:

Alignment vs. accountability. The alignment question - does the model share our values? - is probabilistic and unfalsifiable at the API boundary. The accountability question - can every consequential action be replayed, attributed, and vetoed? - is architectural and receiptable. We argue the industry's trust problem is an accountability problem wearing alignment's clothes.

Evaluation vs. boundary. Benchmarks measure a model; a boundary conditions a model. A 99.9% benchmark and a 90% benchmark look identical from inside a cage that receipts every escape. The evaluation industry's honest frontier is not better scores; it is verifiable operational envelopes.

Determinism vs. usefulness. Caging does not reduce the probabilistic layer to a lookup table. The model retains full creative latitude inside the boundary - proposing, classifying, drafting, discovering. What it loses is unilateral authority over state. In our operating experience this is not a constraint on usefulness; it is the precondition for deploying usefulness where mistakes cost money.

8. Conclusion

A probabilistic language model was placed inside a deterministic mathematical boundary. The boundary caught a real drift event and produced a replayable receipt (Sec. 5). The boundary then caught its own paper fabricating its results (Sec. 6). Both catches are on public record. The cage works, and - more importantly - when it fails, that failure will be receipted too, which is the entire point.

Future work: the 60-day attribution battery (begun October 7, 2026) will produce the labeled dataset the field's evaluation sections pretend they already have. Independent parties are invited to run the falsifier gauntlet. The receipts will be waiting in the ledger.

9. References

(All references are to real, inspectable artifacts. Reference numbering has been checked against the archive; where a reference does not exist, this paper says so rather than inventing one - see Sec. 6.1, item 4.)

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Appendix A: Provenance of This Document

This paper was drafted inside the platform it describes, under bifurcated governance: a probabilistic agent (the author's operator agent) composed the draft; the human author (the deterministic key) reviewed and approved content decisions; the boundary-layer review protocol of Sec. 6 was applied to the draft's claims against the platform's receipt archives. The AI-generated predecessor draft (Sec. 6) is retained, unmodified, in the drift specimen archive. Status of this document: audited draft, pending two-key release. Nothing in this paper is published until the author's explicit release; when published, it will carry a DOI and a ledger entry in that order, consistent with the platform's patent-then-publish doctrine.

End of paper.